Hollow Tube Waveguides

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Lecture Objective

By the end of this lecture, you will:

- Understand the basic features of hollow tube waveguide.

Transmission Lines As TEM Mode Waveguides

- Three canonical transmission lines (properties summarized in Table 2-1):
  - Parallel plate
  - Coaxial cable
  - Two wire
- Transmission lines guide TEM waves.

TEM: $E \perp H \perp k$

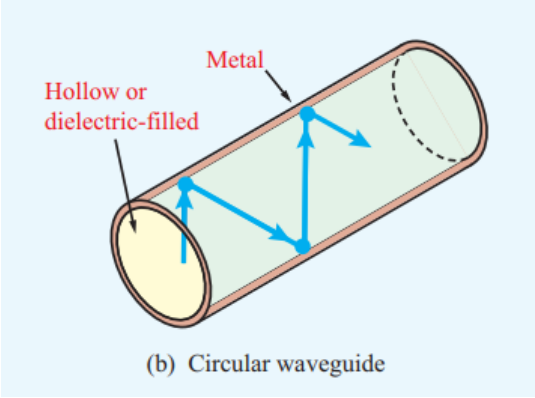
Transmission Lines As TEM Mode Waveguides

- The structures we call “transmission lines” are a subset of a class of structures known as “waveguides”.
- In general, waveguides are structures that support TEM, TE, TM and other types of electromagnetic waves.

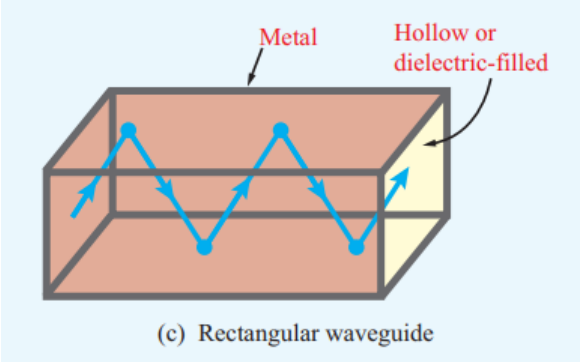
Definitions of the Term “Waveguide”

- The general definition of the term “waveguide” is any structure that guides the propagation of electromagnetic waves.
- The term “waveguide” is also very commonly used to refer to a specific type of waveguiding structure namely the hollow-tube waveguides.

Hollow-Tube Waveguides



(b) Circular waveguide



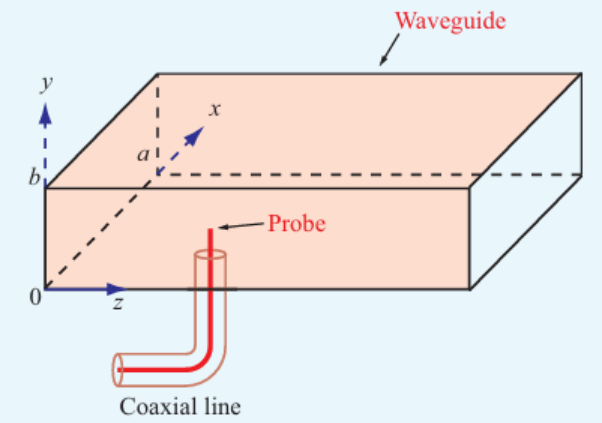
(c) Rectangular waveguide

Do not support TEM modes.

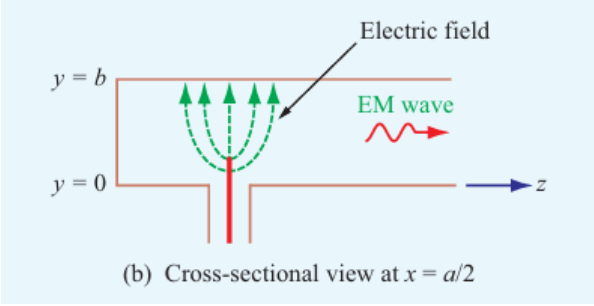
Waveguide Vs. Coaxial Cable

- Coax is the most popular transmission medium; waveguide is significantly more expensive, and bulkier and heavier especially below ~10 GHz.
- At frequencies above ~30 GHz, the dimensions required for the coax to support only TEM mode become prohibitively small.
- Coax's power handling capability is limited by dielectric breakdown; the power handling capability of waveguide is much greater.
- Attenuation due to dielectric loss increases with frequency; air-filled waveguide has much lower loss.

Excitation of Waveguides

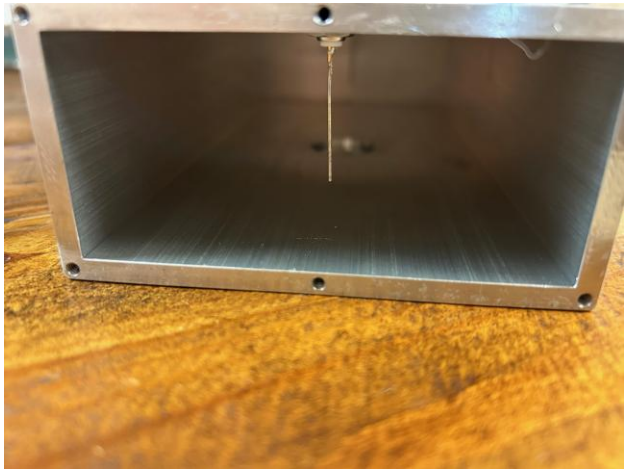


(a) Coax-to-waveguide coupler



(b) Cross-sectional view at $x = a/2$

2. Excitation of Waveguides



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